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12 RADIATION PROTECTION

The purpose of this section is to demonstrate that radiation exposure to plant personnel and persons at the site boundary, from sources contained within the plant and on the site, will be kept as low as practicable and within applicable limits.

12.1 Shielding

12.1.1 Design Objectives

The overall design objectives for shielding during normal operation, maintenance, refueling, and anticipated operational occurrences are:

1. To ensure that external radiation exposure to all onsite personnel remain below the limits set in 10CFR20.

2. To reduce the possibility of radiation induced material damage and potential equipment activation.

In addition, the shielding provided ensures that in the unlikely event of a maximum design accident, the contained activity does not result in any harmful offsite radiation exposures.

All plant areas capable of personnel occupancy are classed as one of the five zones of radiation exposure listed in Table 12.1-1.

Typical Zone I areas are the Turbine Building, and turbine plant service areas and the Control Room. Typical Zone II areas are the outer surfaces of the containment and Auxiliary Building. Zone II areas include the local control spaces in the Auxiliary Building, and the operating deck of the containment during reactor shutdown. Areas designated Zone III include the reactor cavity area after shutdown and the decontamination area. Typical Zone IV areas include the Sampling Room, valve galleries within the Auxiliary Building and areas outside the crane wall of the containment at power operation. Typical Zone V areas are within the regions adjacent to the Reactor Coolant System (RCS) at power operation and the demineralizers and Volume Control Tank Rooms within the Auxiliary Building.

Shielding for the Zone III areas defined above was designed to reduce radiation levels to below 15 mR per hour, and the areas are expected to have a general radiation level that is below 15 mR per hour. However, these areas are classified as Zone IV because small segments of lines (ranging from instrument tubing to 2-inch piping) carrying radioactive liquids penetrate the shield walls, and at contact the dose rates can exceed 15 mR per hour locally, based on the activity associated with 1 percent failed fuel. These small segments of lines are located in such a position that it is not necessary for operating personnel to have contact with them. As such, since these areas are designed for radiation levels of less than 15 mR per hour and fall in the category of periodic occupancy, they can be classified as Zone III areas.

Radiation shielding is designed for operation at maximum calculated thermal power and to limit the normal operation radiation levels at the site boundary to below those levels allowed for

continuous nonoccupational exposure. The plant is capable of continued safe operation with 1 percent fuel element defects.

The shielding is divided into five categories according to function. These functions include the primary shielding, the secondary shielding, the accident shielding, the fuel transfer shielding, and the auxiliary shielding.

All radiation and high radiation areas are appropriately marked and isolated in accordance with 10CFR20 and other applicable regulations except as noted in Section 6.12.1 of the Technical Specifications.

12.1.1.1 Primary Shielding

The primary shielding is designed for the following:

1. Reduce the neutron fluxes incident on the reactor vessel to limit the radiation induced increase in transition temperature.

2. Attenuate the neutron flux sufficiently to prevent excessive activation of plant components.

3. Limit the gamma flux in the reactor vessel and the primary concrete shielding to avoid excessive temperature gradients or dehydration of the primary shield.

4. Reduce the residual radiation from the core, reactor internals and reactor vessel to levels which will permit access to the region between the primary and secondary shields after plant shutdown.

5. Reduce the contribution of radiation leaking to obtain optimum division of the shielding between the primary and secondary shields.

12.1.1.2 Secondary Shielding

The main function of the secondary shielding is to attenuate the radiation originating in the reactor and the reactor coolant. The major source in the reactor coolant is the Nitrogen -16 activity, which is produced by neutron activation of oxygen during passage of the coolant through the core. The secondary shielding will limit the full power dose rate outside the Containment Building from radioactivity inside the containment to less than 1.0 mR per hour.

12.1.1.3 Accident Shielding

The main purpose of the accident shielding is to ensure safe radiation levels outside the Containment Building following a maximum design accident.

12.1.1.4 Fuel Transfer Shielding

The fuel transfer shielding permits the safe removal and transfer of spent fuel assemblies and control rod clusters from the reactor vessel to the spent fuel pool. It is designed to attenuate radiation from spent fuel and control clusters to less than 2.5 mR per hour at the refueling cavity water surface and less than 1.0 mR per hour in the Auxiliary Building.

12.1.1.5 Auxiliary Shielding

The function of the shielding is to protect personnel working near various system components in the Chemical and Volume Control System (CVCS), the Residual Heat Removal (RHR) System, the Waste Disposal System (WDS) and the Sampling System. The shielding provided for the Auxiliary Building is designed to limit the dose rate to less than 2.5 mR per hour in normally occupied areas, and at or below 15.0 mR per hour in periodically occupied areas.

The design criteria for radiation protection in the Auxiliary Building following a loss-of-coolant accident (LOCA) is as follows:

1. The RHR compartment has sufficient shielding to assure that the 8-hour integrated dose outside the RHR compartment will not exceed 3 rem (after the first 24 hours following the accident).

2. The RHR compartment shielding provides for access for manual isolation of pumps.

12.1.2 Design Description

12.1.2.1 Primary Shielding

The primary shielding consists of the reactor internals, the reactor vessel wall, and a concrete structure surrounding the reactor vessel.

The primary shielding immediately surrounding the reactor vessel consists of a reinforced concrete structure extending from the base of the containment to an elevation of 104 feet. The lower portion of the shield is a minimum thickness of 7 feet of concrete and is an integral part of the main structural concrete support for the reactor vessel. It extends upward to join the concrete cavity over the reactor. This cavity is approximately rectangular in shape, and has concrete sidewalls which are 4 feet - 0 inches thick adjacent to areas in which fuel is transported.

The primary concrete shielding is air cooled to prevent overheating and dehydration from the heat generated by radiation absorption in the concrete. Eight "windows" have been provided in the primary shield for insertion of the out-of-core nuclear instrumentation. Cooling for the primary shield concrete, nuclear instrumentation, and vessel supports is provided by circulating 18,000 cfm of containment air between the reactor vessel wall and the surrounding concrete structure.

12.1.2.2 Secondary Shielding

The secondary shielding surrounds the reactor coolant loops and the primary shielding. It consists of interior walls within the Containment Building, the operating floor and the reactor containment building itself. The Containment Building also serves as the accident shield.

The lower portion of the secondary shielding above grade consists of the 4 foot -6 inch thick cylindrical portion of the reactor containment and a minimum of 3 feet thick concrete interior walls surrounding the reactor coolant loops. The secondary shielding will reduce the radiation levels in the primary loop compartment to 15 mrem per hour outside the polar crane wall and to a level of less than 1.0 mR per hour outside the Reactor Containment Building. Penetrations in the secondary shielding are protected by supplemental shielding.

The secondary shielding design parameters are listed in Table 12.1-2.

12.1.2.3 Accident Shielding

The accident shielding consists of the 4 foot - 6 inch reinforced concrete cylinder capped by a shallow, reinforced concrete dome 3 feet - 6 inches thick. Supplemental shielding has been provided for the containment penetrations.

Shielding for the equipment access hatch is credited only for the inner access hatch which is modeled as a 1-1/4" steel plate for all postulated accident scenarios. Smaller penetrations associated with piping and electrical cables are directed into the penetration area which is shielded with a minimum of 24 inches of concrete. The Control Room is protected with concrete sidewalls 24 inches thick, and a concrete roof 24 inches thick.

The accident shielding design parameters are listed in Table 12.1-3.

12.1.2.4 Fuel Transfer Shielding

The refueling cavity is formed by the upper portions of the primary shield concrete, and other sidewalls of varying thicknesses. A portion of the cavity is used for storing the upper and lower internals packages and is shielded with concrete walls 4 feet thick. The remaining walls vary from 4 feet to 6 feet thick, and provide the shielding required for handling spent fuel.

The refueling cavity, flooded with borated water to elevation 128 feet-8 inches during refueling operations, provides a temporary water shield above the components being withdrawn from the reactor vessel. The water height during refueling is approximately 24 feet-8 inches above the reactor vessel flange. This height ensures that a minimum of 10 feet-0 inch of water will be above the top of a withdrawn fuel assembly. Under these conditions, the dose rate is less than 2.5 mR per hour at the water surface.

The spent fuel assemblies and control rod clusters are remotely moved from the reactor containment through the horizontal spent fuel transfer tube and placed in the spent fuel pool. Concrete, 4 feet-6 inches thick on sides and bottom, and 5 feet thick on top shields the spent fuel transfer tube. This shielding is designed to protect personnel from radiation during the time

a spent fuel assembly is passing through the main concrete support of the reactor containment and the transfer tube.

Radial shielding during fuel transfer is provided by the water and concrete walls of the fuel transfer canal. An equivalent of 6 feet of concrete is provided to insure a maximum dose value of 1.0 mR per hour in the Auxiliary Building areas adjacent to the spent fuel pool from a spent fuel assembly in the fuel transfer tube.

Spent fuel is stored in the spent fuel pool which is located adjacent to the Containment Building. Radial shielding for the spent fuel is provided by 6-foot thick concrete walls plus a minimum of 10 1/2 inches of water. The pool is flooded with borated water to a level such that the water height above the stored fuel assemblies is approximately 25 feet.

The refueling shielding design parameters are listed in Table 12.1-4.

12.1.2.5 Auxiliary Shielding

The auxiliary shielding consists of concrete walls around certain components and piping which process reactor coolant. In some cases, the concrete block walls are removable to allow personnel access to equipment during maintenance periods. Each equipment compartment is individually shielded so that compartments may be entered without having to shut down and, possibly, to decontaminate the adjacent system.

The shielding material provided throughout the Auxiliary Building is ordinary concrete, with some lead and steel supplemental shielding. The principal auxiliary shielding provided is tabulated in Table 12.1-5.

12.1.3 Source Terms

The residual heat removal loop radiation sources and evaluation parameters are developed in Section 15.4.1. The radiation sources which are assumed to be released to the containment following a DBA LOCA with fuel failure per Regulatory Guide 1.183 are developed in Section 15.4.1.

12.1.3.1 Miscellaneous Materials

The Salem operating licenses authorize PSE&G to receive, possess, and use in amounts required any byproduct, source, or special nuclear material without restriction to chemical or physical form, for sample analysis, instrument calibration, or other uses associated with radioactive apparatus or components. This authorization is provided pursuant to 10CFR Parts 30, 40, and 70.

Radioactive sources are inventoried and leak checked as required by Technical Specification 3/4.7.8.

A source locker under the control of Radiation Protection provides secure storage and administrative control for licensed and non-licensed sources. Sources may be stored and

secured in other station locations if they are physically too large for the source locker or if their presence in the source locker would present an unnecessary exposure hazard or if the source/sources is/are installed as part of a system such as the RMS or for interim storage as directed by RP.

12.1.3.1.1 Primary and Secondary Neutron Sources

Neutron source assemblies may be utilized in the core. These will consist of assemblies containing primary and secondary source rods. The rods in each assembly will be fastened to a hold down plate at the top end. The hold down plate is similar to that of the burnable poison and plugging device assemblies. The secondary sources (Sb-Be) are initially nonradioactive, becoming a neutron source only after activation in the core. The secondary source rods, 0.381 inch in diameter contain Sb-Be pellets with an overall height of 88 inches. Primary source rod assemblies contain capsules of Pd-$Cf_2 0_3$ source material 1.5 inches long and alumina rods to space the capsule and fill the remainder of the rod height. Primary source assemblies are used for initial cycle loading and as required after long shutdown when secondary source strength has decayed to unacceptably low levels.

12.1.3.1.2 Primary Source Rod Assembly Data

Primary Source used in Unit 1 Cycle 1 and Unit 2 Cycle 1 loadings:

Type - Triple encapsulated Californium Pd-$Cf_2 0_3$ cermet

Mfg. - Gamma Industries - Division of Nuclear Systems

Neutrons/sec. - Approximately 4×10^8

Size - Approximately 1.5 inches long - 0.381 inch OD same as fuel rod

Encapsulation:

Primary - Savannah River National Laboratories, Type SR-CF-lx

Intermediate - Savannah River National Laboratories, Type SR-CF-100

Outer - Same design as fuel rod

Isotope	WT (%)
Cf-252	79
Cf-251	4
Cf-250	15
Cf-249	2

Shipment - Common carrier using Gamma Industries container SISC-1, DOT-Type A-7A

Primary Source used in Salem Unit 1 Cycle 13 loading:

Type-	Triple encapsulated Californium capsule
Mfg.-	General Electric - Vallecitos Nuclear Center
Neutrons/sec-	Approximately 7.5 x 10^8
Adapter Body Size-	Approximately 1.5 inches long - 0.330 inch ID and OD same as fuel rod
Encapsulation:	
Primary-	Welded inner capsule body, Special Form Certificate Number USA/0141/S, Rev.8 Model GEN-CF-1x
Intermediate- (Adapter Body)	Inner capsule body inserted into an adapter body and closed by a press-fit inner plug
Outer-	Same design as fuel rod

Isotope	Wt(%)
Cf-252	80.282
Cf-251	3.144
Cf-250	10.342
Cf-249	6.227

Shipment-	MRC Model 2511-C Radioactive Shipping Container, U.S.D.O.T. Specification 7A Certification

12.1.3.1.3 Reactor Vessel Flux Dosimeters

As part of the reactor vessel surveillance program, a number of dosimeters are located in specimen capsules which are located about 3 inches from the vessel wall directly opposite the center portion of the core. The dosimeters permit evaluation of the neutron flux seen by the specimens and vessel wall.

Number of dosimeters	11-Np-237 11-U-238
Total amount of contained NP-237	197 mg
Total amount of contained U -238	132 mg
Length of encapsulation material, in.	0.375

Diameter of encapsulation material, in.	0.25
Material used to encapsulate U-238	Brass
Material used to encapsulate NP-237	Stainless Steel

Note: Eight material test capsules were inserted in Salem Units 1 and 2 reactor pressure vessels prior to initial plant startup. Salem Unit 1 contained 5 Type I and 3 Type II capsules. Salem Unit 2 was provided with all 8 Type II capsules. Only Type II capsules contain NP-237/U-238 dosimeter blocks. Hence, a total of eleven dosimeters were loaded in Salem Units 1 and 2 reactor pressure vessels. To date, 2 capsules have been removed from each of the Salem Units (i.e., 1 Type I and 1 Type II capsule from Salem 1; 2 Type II capsules from Salem 2).

12.1.3.1.4 Special Nuclear Materials (SNM)

The Salem Operating License authorizes PSE&G to receive, possess and use any SNM as reactor fuel and as fission detectors in any amounts as required. This includes, but is not limited to, incore monitoring fission chamber detectors and excore fission chamber detectors. These detectors are considered "Licensed Material" in accordance with (January 1995) 10CFR20.1003 Definitions. These detectors are also considered "Sealed Sources" in accordance with (January 1995) 10CFR74.4.

Special nuclear material in the form of reactor fuel will be received, stored, used and shipped in the normal operation of the nuclear plant. The facilities provided to ensure safety during these transfer and storage operations are consistent with the as low as practicable exposure guidelines.

A separate Fuel Handling Building is provided for each nuclear unit. These buildings are provided with fire detection systems, flood protection, radiation monitoring systems and facilities for cask decontamination. During fuel movement, additional equipment for radiation monitoring will be available in these areas. Portable survey instruments for radiation dose rate measurements, contamination surveys and air activity measurement will be provided during these periods. Protective clothing will be provided to personnel in these areas as conditions require.

Special Nuclear Material in the form of incore monitor fission detectors is used to monitor the neutron flux distribution within the reactor core. These detectors are sealed units that contain Special Nuclear Material (SNM) enriched in U-235. Unirradiated detectors will be stored in a secure area which prohibits unauthorized removal or access in accordance with (January 1995) 10CFR20.1801, "Security of Stored Material." Irradiated detectors which have been removed from service may be temporarily stored in the Seal Table Room shield wall pipes. After the detectors have decayed to acceptable levels for handling and movement in accordance with radiation protection practice (ALARA), the detectors will be removed from the shield wall pipes and stored in an acceptable radiation storage area which limits access and prohibits unauthorized removal.

Special Nuclear Material in the form of excore fission chamber detectors are used as an independent system to monitor the neutron flux level from shutdown to full power. This system is required for post accident monitoring and for remote shutdown monitoring capability. These detectors are sealed units that contain Special Nuclear Material (SNM) enriched in U-235. Irradiated excore fission detectors will be stored in a locked high radiation area. Unirradiated detectors will be stored in a secure area which prohibits unauthorized removal or access is accordance with (January 1995) 10CFR20.1801, "Security of Stored Material."

12.1.3.1.5 Fuel and Fuel Handling

Shipment of the fuel assemblies from the fabricating plant to the Fuel Handling Building will be by truck trailers. Not more than 16 fuel assemblies will be delivered in any one shipment. The new fuel assemblies will be removed one at a time from the shipping containers by the fuel handling crane and will be checked visually for integrity and numbering and surveyed for background radiation levels and possible contamination.

All activities of unloading, inspecting, moving, removing, and replacing the fuel assemblies will be performed in accordance with written procedures.

Assemblies will not be handled during severe weather conditions; for example, during periods when a hurricane or tornado watch is in effect in the vicinity of the site.

A Geiger-Mueller, or equivalent, monitor is located on the operating deck floor (Elevation 130 feet) of each Fuel Handling Building. These monitors are described in Section 11.4.2.4. In lieu of maintaining a criticality monitoring system as described in 10 CFR 70.24, Salem complies with the requirements of 10 CFR 50.68(b).

In addition, personnel authorized to enter the Fuel Handling Building shall carry personnel radiation monitoring devices. Radiation monitoring of the facility area shall be conducted on a routine basis, and portable radiation survey instruments shall be used during all fuel handling operations.

Comprehensive emergency procedures have been written to ensure that all personnel withdraw upon the sounding of the alarm to a designated area of safety. Procedures are approved by the Plant Manager and include provisions for instruction of personnel and conduct of drills to familiarize them with the evacuation plan. Portable radiation survey meters are in accessible locations for use in such an emergency.

Before the receipt of new fuel, all personnel associated with the fuel handling operation will have received training in Health Physics and fuel handling procedures.

12.1.4 Area Monitoring and Radiation Surveys

The primary purpose of the area monitors is to prevent excessive personnel exposure by monitoring locations within the plant. A secondary purpose is for emergency response assessment and to provide radiation data in support of the as low as is reasonably achievable (ALARA) program and Regulatory Guide 8.8. Area monitors are located in the station

radiologically controlled area where dose rates may be used to indicate potentially degrading plant conditions. The monitors provide local indication and alarms and operate horns and flashing beacons upon high radiation conditions. Control Room indications and alarms are also provided. The area monitoring system is described in Section 11.4. Onsite areas external to plant buildings are monitored by thermoluminescent dosimeters.

Radiation surveys are conducted to identify and control radiation sources associated with the operation of the Salem station. They are also performed to mark and verify radiation area boundaries, and to determine if abnormal radiation levels exist. The location and frequency of these routine radiation surveys are defined in the station Radiation Protection procedures. Radiation, High Radiation, and Very High Radiation Areas are posted conspicuously to ensure compliance with 10CFR20. Radiation survey instruments are calibrated using current industry standards and guidelines.

12.1.5 Estimates of Exposure

The average annual onsite person-rem exposure resulting from plant operations is approximately 409 person-rem per year for Unit 1. This is based on operational data from 1976 through 1982 with values ranging from a low of 117 person-rem in 1977 to a high of 1027 person-rem in 1982, which included two refueling outages.

The average annual onsite person-rem exposure resulting from plant operations is approximately 16 person-rem per year for Unit 2. This is based on data for 1981-1982, during which time no major outages occurred.

Annual dose goals based on scheduled maintenance and/or refueling activities are established by station management.

Zone	Condition of Occupancy	Maximum Dose Rate (1% failed fuel mrem/hr
I	Unlimited Occupancy	Less than 0.25
II	Normal Continuous Occupancy	0.25 - 2.5
III	Periodic Occupancy	2.5 - 15.0
IV	Controlled Occupancy	15.0 - 100
V	Controlled Access	Greater than 100

Core power density, w/cc	109
Reactor coolant liquid volume, ft^3	12,612
Reactor coolant transit times:	
Core, sec	0.796
Core exit to steam generator inlet, sec	2.111
Steam generator inlet channel, sec	0.700
Steam generator tubes to vessel inlet, sec	6.075
Vessel inlet to core, sec	2.560
Total out of core, sec	12.242
Full power dose rate outside secondary shield, mR/hr	1.0

Core thermal power	3632 MWt
Minimum full power operating time	1000 days
Equivalent fraction of core failed	1.0
Fission product fractional releases:	
Noble gases	1.0
Halogens	0.4
Alkali Metals	0.3
Containment air cleanup rate by CFCU	
Filtration following accident	0.0
Maximum 30-day accident dose	
Through containment shielding (due to containment shine) in Control Room, rem TEDE	< 1.0

Total number of fuel assemblies	193
Minimum full power exposure	1000 days
Minimum time between shutdown and fuel handling	56 hours
Minimum dose rate adjacent to spent fuel pool	1.0 mR/hr
Maximum dose rate at water surface	2.5 mR/hr

Component	Concrete Shield Thickness, ft-in.
CVCS demineralizers	3 – 0 to 4 – 6
Other demineralizers	1 – 0 to 4 – 0
Charging pumps	1 – 6 to 2 – 6
Liquid holdup tanks	2 – 0 to 3 – 9
Volume control tank	3 – 6
Reactor coolant filter	3 – 6
Boric acid evaporator package	2 – 0
Gas decay tanks	2 – 6 to 3 – 9
Gas compressor	2 – 6 to 3 – 9
Waste evaporator	2 – 0
Liquid waste holdup tank	2 – 0
Spent resin storage tank	3 – 6 to 4 – 0

Design parameters for the auxiliary shielding include:

Core thermal power	3558 MWt
Fraction of fuel rods containing small clad defects	0.01
Reactor coolant liquid volume	12,612 ft^3
Letdown flow (normal purification)	75 gpm
Effective Cesium purification flow (intermittent)	7.5 gpm
Dose rate outside auxiliary building	< 2.5 mR/hr
Dose rate in the building outside shielded compartments	< 2.5 mR/hr

12.2 Ventilation

12.2.1 Design Objectives

The plant Ventilation Systems are designed to provide safe atmospheric conditions within the plant at all times. Design objectives include, among others, limiting the spread of airborne radioactivity.

12.2.2 Design Description

The description of each Ventilation System is given in the respective sections as follows:

Section No.	Title
6.2.2.2	Containment Fan Cooling System
9.4.1	Control Area Air Conditioning System
9.4.2	Auxiliary Building Ventilation System
9.4.3	Fuel Handling Area Ventilation System
9.4.4	Containment Ventilation System
9.4.5	Diesel Generator Area Ventilation System
9.4.6	Switchgear Room Ventilation System
9.4.7	Service Water Intake Structure Ventilation System

12.2.2.1 Equipment Sizing

Flow rates are given in the Equipment Data Tables in terms of total flow and on the basis of air changes for a given space per hour.

All exhaust systems are designed on the basis of providing full air flow capacity at the maximum static pressure losses associated with dirty filters just prior to servicing.

12.2.2.2 Filter Characteristics

Filter characteristics are given in the Equipment Data and Materials of Construction Tables in respective sections.

12.2.2.3 Post Accident System Operation

Post accident shutdown and emergency mode operation is described in the respective sections.

12.2.3 Radiation Monitoring

The provisions for plant radiation monitoring are described in Section 11.4.

12.3 Radiation Protection Program

12.3.1 Program Objectives

The Radiation Protection Program provides evaluation and documentation of site radiological conditions and ensures that every reasonable effort is made to maintain personnel exposures "as low as reasonably achievable" (ALARA) in accordance with requirements of 10CFR20, Regulatory Guides, and Technical Specifications. The program is designed to protect the public and plant personnel from unnecessary exposure to radiation and radioactive materials. The personnel responsible for the Radiation Protection Program are, in order of authority, the Salem Site Vice President, the Plant Manager, the Radiation Protection Manager, Radiation Protection Supervisors, and Radiation Protection Technicians.

12.3.2 Organization

The Plant Manager is responsible for maintaining and implementing the Radiation Protection Program and receives direct reports from the Radiation Protection Manager concerning the status of the program.

The Radiation Protection Manager (RPM) is responsible for managing the Radiation Protection department to meet station operational needs and radiological safety standards. The Radiation Protection Manager manages the Radioactive Material Control program and implements the ALARA program as described in administrative procedures.

Radiation Protection Supervisors are responsible for planning, conducting, and supervising daily radiation protection activities.

Radiation Protection Technicians implement the radiation protection program under the supervision of Radiation Protection Supervisors.

Radiation Protection personnel have the authority to halt any work activity when, in their professional judgment, worker safety is being jeopardized or unnecessary personnel exposures are occurring.

In the absence of Radiation Protection supervision, the authorities of the above positions may be delegated in accordance with station radiation protection procedures to qualified supervisors or technicians.

The Radiation Protection Manager is familiar with the design features of nuclear power stations and possesses both the technical competence to establish radiation protection programs and the supervisory capability to direct the work of the professionals and technicians required to implement such programs.

The qualifications of the designated "Radiation Protection Manager" meet or exceed the requirements of Regulatory Guide 1.8, September 1975.

At least one member of the Radiation Protection Supervisor staff shall be designated as the backup “Radiation Protection Manager” in accordance with paragraph 4.4.4(d) of ANSI/ANS 3.1-1981.

The Radiation Protection Supervisors are qualified in accordance with ANSI/ANS 3.1-1981. They shall have a minimum of four years of experience in applied radiation protection, including two years of experience in a nuclear power plant or a nuclear facility.

The qualifications of the Radiation Protection Technicians meet or exceed the personnel requirements of ANSI/ANS 3.1-2014. Radiation Protection Technicians are additionally trained and qualified in accordance with administrative procedures.

12.3.3 Facilities, Equipment and Instrumentation

12.3.3.1 Personnel Protective Equipment

Protective Clothing

Personnel entering the radiologically controlled area (RCA) are required to wear protective clothing in contaminated areas. The nature of the work as well as the levels of contamination will govern the selection of protective clothing to be worn by individuals.

Protective apparel available is provided by Radiation Protection in accordance with plant conditions and need including items of specialized apparel such as plastic or rubber suits, face shields, and respirators are available for operations involving high level contamination. In all cases, radiation protection personnel will evaluate the radiological conditions and specify the required items of protective clothing to be worn on the Radiation Work Permit (RWP).

Respirators

Respiratory protective devices may be used in situations where airborne radioactivity which cannot be mitigated by engineering controls exists or is expected. In such cases, the airborne concentrations are monitored by radiation protection personnel. Protective devices, required according to concentration and type of airborne contaminants present, are specified on the RWP.

Self-contained breathing apparatus are available for use in situations involving exposure to significant gaseous activity or an oxygen deficient atmosphere.

The use of Delta Protection Mururoa V4 F1 and V4 MTH2 respiratory protection suits has been authorized for use at Salem with an assigned protection factor (APF) of 2,000 (Reference NRC to PSEG letter: “HOPE CREEK GENERATING STATION AND SALEM NUCLEAR GENERATING STATION, UNIT NOS. 1 AND 2 -REQUEST FOR AUTHORIZATION TO USE RESPIRATORY PROTECTION EQUIPMENT (TAC NOS. MD9199, MD9200, AND MD9201)”, dated January 27, 2009). Approval was based on testing which demonstrated the suits met European Standard EN 1073-1 (January 1998), "Protective Clothing Against Radioactive Contamination, Part 1: Requirements and Test Methods for Ventilated Protective Clothing

Against Particulate Radioactive Contamination," This standard is generally consistent with the pertinent acceptance criteria provided in Los Alamos National Laboratory Report LA-10156-MS, which is used to test and authorize the use of air-supplied suits at Department of Energy sites. The certification-testing was broadly based covering a range of various functional areas. Both models passed all required tests, and both provided a measured average protection level (fit factor) of 50,000. The following information on the Mururoa V4 F1 and V4 MTH2 suits is included to comply with commitment CM.CC.2008-121:

The manufacturer’s instructions for use and storage of the Delta Protection Mururoa V4F1 and V4 MTH2 suits will be adhered to and integrated into the respiratory protection program, with the exception of the requirement to have a stand-by rescue person. New lesson plans will be developed to train workers on Mururoa's features, donning, use and removal, cautions and use of mouth strip and tear off strips for routine and emergency egress. Radiation Protection personnel will be provided additional training for selection, approval, issue, equipment set-up, operation and maintenance instructions for the Mururoa suit. The Mururoa V4F1 and V4 MTH2 suits will be discarded after a single use and will not be used in atmospheres that are immediately dangerous to life and health (IDLH). Any defects discovered with the Mururoa suit will be entered into the Corrective Action Program and reported to the manufacturer, as necessary. Industry notifications, when required, will be made through the Operating Experience Program.

12.3.3.2 Facilities

The general arrangement of the service facilities is designed to provide adequate personnel decontamination and change areas and to control access to the Radiologically Controlled Area (RCA).

The male and female clean locker room is used to store items of personal clothing not required or allowed in the radiologically controlled area and is employed as a protective clothing change area. A supply of clean protective clothing for personnel is maintained in this area.

All personnel will survey themselves before leaving the radiologically controlled area utilizing an approved personnel monitoring device provided by radiation Protection. Additional radiation monitoring devices for frisking tools, etc. are also available at the access control point.

A decontamination area is provided within the radiologically controlled area for the decontamination of hand tools and small equipment.

Radiation protection facilities include equipment to count air samples and contamination wipe surveys. A sufficient quantity of each type of instrument is available to support both routine operations and emergency response. The selection of radiation detection instrumentation is based on the ability of the instrument to perform with reliability and accuracy. Each instrument chosen for use is evaluated on the following parameters:

a. Ease of calibration and repair;

b. Interchangeability of components;

c. Weight and size for user acceptance;

d. Standard readouts;

e. Response to radiation of interest.

Outside laboratories may be used for independent analysis or special samples beyond the scope of the stations' capabilities.

12.3.3.3 Area Control

The plant site is divided into two general areas: the protected area and the owner-controlled area. The protected area is that portion of the site which is enclosed by fencing and is posted with "No Trespassing" signs. The owner-controlled area encompasses all areas external to the fence but within the site property line.

The restricted area for the Salem Generating Station, as defined by 10CFR20.3, begins at the security fence (i.e., the protected area). Access to the restricted area is controlled by the licensee for the purposes of protection of the individuals from exposure to radiation and radioactive material. The unrestricted area as defined by 10CFR20.3 includes all areas outside the security fence which are not controlled by the licensee for the purposes of protection of individuals from exposure to radiation and radioactive material. The Radiologically Controlled Area (RCA) consists of discrete areas within the restricted area which are posted to alert individuals to hazards or potential hazards due to the presence of radiation or radioactive material. Salem station implements certain administrative controls for entrance to and exit from the RCA for the purposes of enhancing and simplifying radiological safety.

Access to the RCA is limited to those persons authorized for entry by plant supervisors and radiation protection personnel through the use of RWPs.

Any area within the RCA containing radioactive materials and radiation is surveyed, classified, and conspicuously posted with appropriate radiation caution signs.

Administrative and physical measures are employed to prevent unauthorized or unintentional entry of personnel into any High Radiation Area or Very High Radiation Area. Alarms, barricades, and locked doors are employed as necessary to restrict access to, and provide warnings of High Radiation Areas or Very High Radiation Areas.

One point (control point) of access to and exit from the RCA is normally utilized. It is located at Elevation 100 feet (ground level) in the Service Building. Other access points may be established as required by plant conditions.

The facilities are designed to provide proper radioactive material and contamination control and are common to both units.

12.3.4 Personnel Dosimetry

Individual dose monitoring devices are normally provided for all personnel who enter the RCA. As a minimum, personnel monitoring consists of a self-reading dosimeter (electronic dosimetry or another direct-reading dosimeter). Individuals who require monitoring in accordance with 10 CFR 20.1502 are normally also monitored with dosimeters of legal record (DLR). Individuals who do not enter the RCA as a part of their jobs are not normally provided with individual dose monitoring devices.

Official record dose for individuals who require monitoring in accordance with 10 CFR 20.1502 is usually obtained from DLR readings. Radiation Protection may specify a different source as a basis for determination of record dose.

Monitoring in accordance with 10 CFR 20.1502 is usually required only for deep dose equivalent (DDE) and lens dose equivalent (LDE). The individual monitoring devices previously discussed are normally used to monitor DDE and LDE. The air sampling program and the bioassay program (principally whole body counting) are used to show compliance with the monitoring thresholds for committed dose equivalent (CDE) and committed effective dose equivalent (CEDE). Passive techniques such as whole body contamination monitors at the RCA exit assist in showing compliance. Individuals who work in the RCA normally receive whole body counts upon initial entry, annually while employed, and upon termination, depending upon the potential for detectable radioactivity. Radiation Protection may prescribe other methods for demonstrating compliance as appropriate.

12.3.4.1 NVLAP Certified Dosimetry

Occupational whole body beta and gamma radiation exposure in the RCA is normally monitored by NVLAP certified Dosimetry for individuals who require monitoring in accordance with 10 CFR 20.1502. Occupational whole body neutron radiation exposure is determined whenever significant neutron exposure is likely. It is determined either by calculation, based on area stay times and survey or neutron/gamma ratio information, or by neutron-sensitive dosimeters.

Special or additional DLRs are issued at the discretion of radiation protection personnel as dictated by working conditions or the RWP.

Dosimeters are processed by an appropriate approved vendor on a periodic basis. Approved vendors of dosimetry services are required to maintain current accreditation by the National Voluntary Laboratory Accreditation Program in all applicable areas.

12.3.4.2 Self-Reading Dosimeters

A self-reading dosimeter (SRD) is issued to all individuals working inside the RCA to provide a continuous, real-time indication of accumulated dose during each individual entry. Electronic dosimeters are normally used. SRDs are read regularly and recorded upon exit from the RCA. SRDs furnish the exposure data for the administrative control of external gamma radiation exposure. Each individual is required to examine his/her SRD reading frequently while in radiation areas.

Self-reading dosimeters are tested and calibrated in accordance with current industry standards.

12.3.4.3 Administrative Exposure Control

Radiation Work Permits (RWPs) are required for work in areas where excessive exposure to personnel is possible. Radiation Work Permits describe the work to be performed and specifies special precautions, and protective clothing.

PSE&G has established an incremental series of administrative controls to carefully monitor radiation exposures. The Radiation Protection Program is the governing document which defines these controls. The administrative controls ensure that under normal operating conditions no regulatory limits will be exceeded. These administrative limits are below the federal limits for occupational radiation exposure in 10CFR20.

Processing of dosimetry devices is carried out as required to maintain a record of cumulative exposure. In addition, special processing will be initiated whenever it is suspected that the individual has received a cumulative dose approaching the maximum annual limit.

12.3.5 Procedures

12.3.5.1 Departmental Procedures

Radiation Protection procedures have been developed in accordance with station procedures to provide administrative control over radiation protection areas of responsibility, including:

a. Organization, Administration, Training and Qualification;

b. Access Control;

c. External Exposure Monitoring;

d. Internal Exposure Monitoring;

e. Instrumentation and Sources;

f. Surveys;

g. Dose Control;

h. Radioactive Material Control;

i. Shipment and Receipt of Radioactive Material;

j. Radiological Incidents;

k. Special Evolutions.

12.3.5.2 Access Training

All personnel requiring unescorted access to the stations are trained in the basic principles and practices or radiation protection. The level of training required is commensurate with the individual's responsibilities. Successful completion of a written exam is required at each level of training. This training requirement does not apply to authorized individuals who have completed an approved training program elsewhere. Requalification is required on a routine basis. Visitors requiring escorted access are provided a briefing on radiation protection principles and practices appropriate to the circumstances of the visit.

12.4 ALARA Program

PSEG Nuclear and its management are committed to operating the Salem Generating Station in a manner that will provide radiation protection for their employees, contractors, and the public. The ALARA program instituted by this commitment will set radiation exposure standards at the station well within federal regulatory limits. Training emphasizes the company commitment to the ALARA philosophy and the individual's responsibilities to that philosophy.

The President and Chief Nuclear Officer has the corporate responsibility for establishing the ALARA program. The responsibility for coordination and administration of the corporate ALARA program is delegated to the Radiation Protection Manager (RPM) who ensures that the policies and commitments contained in the ALARA program are properly implemented.

The RPM ensures proper implementation of the ALARA Program by:

1. Developing policies for the ALARA Program.
2. Ensuring radiation protection policies support the ALARA Program.
3. Ensuring periodic reviews of the ALARA Program are conducted.
4. Reviewing the effectiveness of radiation protection and ALARA training programs.

The Plant Manager is responsible for ensuring the support of all station personnel for the ALARA program and directs the development of administrative and operating procedures designed to guide personnel in maintaining their exposures ALARA.

The ALARA program is supported by the Radiation Protection Manager. The Radiation Protection Manager's major ALARA responsibilities include:

1. Reviews of design changes for facilities and equipment that can affect radiation exposures;
2. Identifying location, operations, and conditions that have the potential for causing significant exposures to radiation;
3. Assisting station management in the establishment of person-rem goals;
4. Reviewing applicable station departmental procedures to maintain exposures ALARA;
5. Tracking exposures of station personnel by specific job functions and type of work.